Bangla Student Feedback Analysis Using Natural Language Processing

Student Research Proposal • Najmus Sakib Yeamin • Natural Language Processing • Data Science • Machine Learning

Proposed by	Najmus Sakib Yeamin
Project type	Student-led research / prototype study
Estimated duration	8–12 weeks

Project Overview

This project explores whether Bangla student feedback can be organized and summarized automatically using natural language processing. The goal is not to replace human interpretation, but to create a simple research prototype that can group feedback by topic and identify broad sentiment or recurring concerns.

Student perspective
The project is intentionally scoped so that a small student team can learn, experiment, and produce a useful research artifact without assuming access to a large laboratory or proprietary dataset.

Research Problem

Student feedback is often collected as free-text comments, which makes it difficult to review a large number of responses consistently. Bangla text adds challenges such as spelling variation, informal writing, mixed Bangla-English sentences, and limited labeled resources. A small, carefully designed study could show which preprocessing and modeling choices are useful for this setting.

Research Objectives

- Build a small, ethically collected and anonymized Bangla feedback dataset.
- Develop preprocessing steps for Bangla text, including normalization and removal of irrelevant tokens.
- Identify common feedback topics such as teaching, course content, assessment, facilities, and support.
- Compare a simple traditional ML classifier with a lightweight deep-learning approach where resources allow.
- Analyze common errors caused by spelling variation, code-mixing, short comments, and ambiguous language.

Research Questions

RQ1. Which preprocessing steps improve the consistency of Bangla feedback?

RQ2. Can a simple TF-IDF classifier provide a useful baseline?

RQ3. Which topics occur most frequently in student feedback?

RQ4. What types of Bangla comments are hardest for automated models to classify?

Proposed Methodology

Step 1. Design a simple annotation guideline and collect only voluntary, anonymized feedback for the study.

Step 2. Perform text cleaning, normalization, tokenization, and exploratory analysis.

Step 3. Create baseline features such as TF-IDF and evaluate models such as logistic regression or linear SVM.

Step 4. Optionally compare the baseline with a suitable pretrained language model or embedding approach.

Step 5. Report accuracy, precision, recall, F1-score, topic distribution, and qualitative examples of model errors.

Data / Materials / Experimental Resources

The exact resources will be finalized with the collaborator after a feasibility discussion. The preferred approach is to start with public or self-generated material, keep the experiment small enough to reproduce locally, and record the source, preprocessing decisions, assumptions, and version of each important component.

Initial resource plan: Anonymized Bangla text, annotation guidelines, preprocessing scripts, model evaluation notebook.

Evaluation Plan

Evaluation will be based on measurable criteria rather than only a successful demo. Depending on the project, this may include classification metrics, detection quality, latency, error analysis, reproducibility, or qualitative review. Baselines will be established before more complex techniques are introduced so that the contribution of each change can be discussed clearly.

Reproducibility and Documentation

The team will keep a simple experiment log containing dataset versions, preprocessing choices, model or system settings, evaluation results, and important observations. Code will be organized so that the main experiment can be rerun without relying on undocumented manual steps. Charts and tables will be generated from saved results where possible.

Expected Challenges

- Limited time and computing resources may require a smaller experimental scope.
- Dataset quality may affect conclusions more than model complexity.
- A strong-looking metric may hide weaknesses that appear during error analysis.
- The research question may need refinement after an initial feasibility experiment.

Planned Mitigation

The project will begin with a small baseline and a feasibility check. If data or hardware limitations make the original scope unrealistic, the team will narrow the research question rather than artificially inflating results. Each limitation will be reported as part of the final research discussion.

Expected Outcomes

- An anonymized Bangla feedback dataset with a transparent annotation process.
- A reproducible preprocessing and classification pipeline.
- Topic and sentiment analysis results with an error analysis.
- A discussion of the practical limitations of NLP for informal Bangla feedback.

Potential Research Collaboration

I would value a collaborator interested in Bangla NLP, annotation quality, or educational data. Someone with experience in language modeling could also help evaluate whether a more advanced approach is justified for a small student-built dataset.

Technical Stack

Python, pandas, scikit-learn, NLP libraries, Jupyter Notebook; optional PyTorch/transformers.

Scope, Ethics, and Responsible Research

Only voluntary and anonymized feedback will be used. Names, student IDs, contact details, and other unnecessary identifiers will not be included. The analysis will remain a research prototype rather than an automated decision-making system.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Dataset/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, implementation, experiment design, data analysis, or writing. I would prefer to divide work according to each person's strengths and keep decisions, code, data, and results documented.

Possible Next Step

If a researcher is interested, I would like to first discuss the research question, available resources, and a realistic scope. After that discussion, the proposal can be refined into a clearer study plan with specific responsibilities and milestones.

Note: This document describes a proposed student project. It does not claim completed research, published results, or guaranteed performance. The scope is open to refinement through collaboration.